

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
```

```
In [3]: df = pd.read_csv(r"C:\Users\Vinay Kumar\Downloads\archive (3)\student_placement_
df.head(5)
```

```
Out[3]:
```

	Age	Gender	College_Tier	Specialization	CGPA	DSA_Problems_Solved	Internships
0	19	Male	Tier 3	AI/ML	7.70	229	0
1	21	Other	Tier 3	Civil	6.97	486	0
2	20	Other	Tier 2	Electronics	6.74	180	0
3	18	Other	Tier 2	Computer Science	6.64	98	4
4	18	Other	Tier 2	Electronics	8.67	216	4



```
In [4]: df.shape
```

```
Out[4]: (20000, 20)
```

```
In [6]: df.dtypes
```

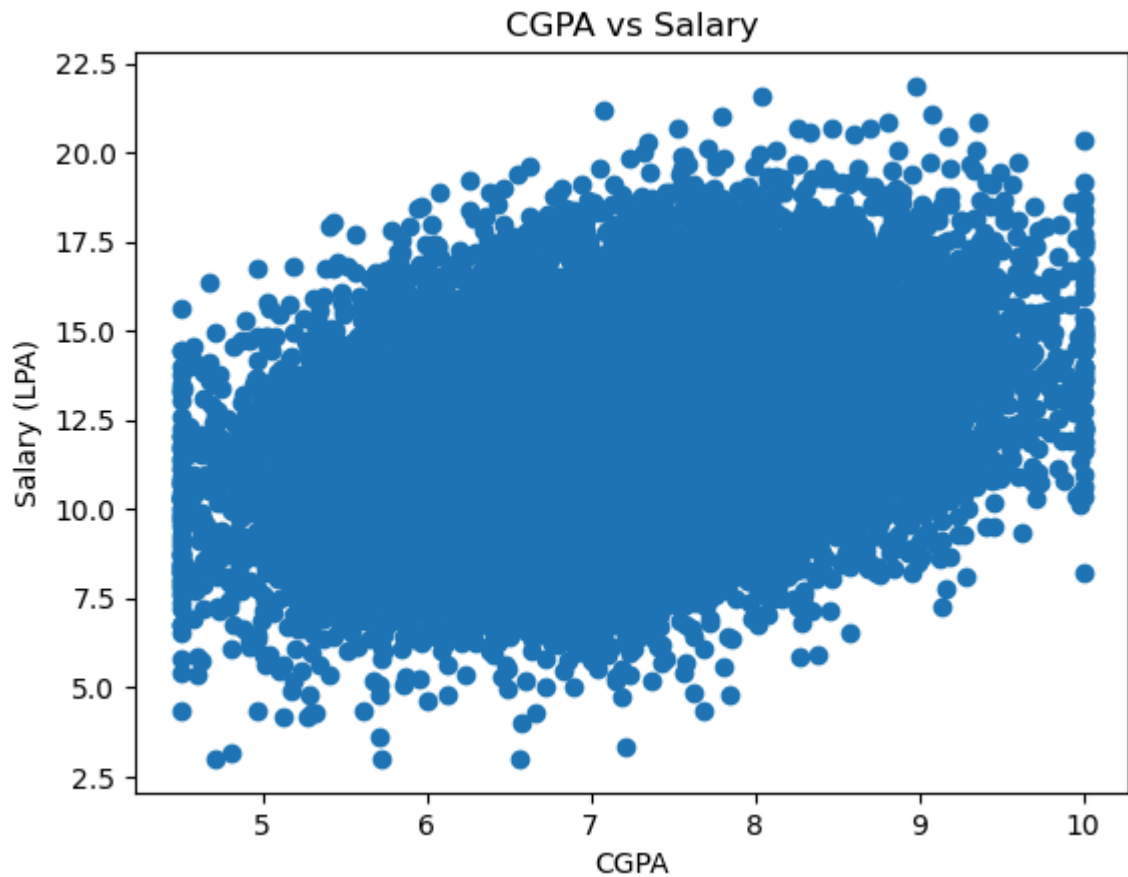
```
Out[6]: Age                int64
Gender                object
College_Tier          object
Specialization         object
CGPA                  float64
DSA_Problems_Solved    int64
Internships            int64
Certifications         int64
Projects_Count         int64
Communication_Skills    int64
Aptitude_Test_Score     int64
LeetCode_Rating         int64
GitHub_Contributions    int64
Hackathons_Participated int64
AI_ML_Skill_Level       int64
System_Design_Knowledge int64
Resume_Score            int64
Mock_Interview_Score     int64
Placement_Status        int64
Salary_LPA             float64
dtype: object
```

```
In [8]: import matplotlib.pyplot as plt

plt.scatter(df['CGPA'], df['Salary_LPA'])

plt.title("CGPA vs Salary")
plt.xlabel("CGPA")
plt.ylabel("Salary (LPA)")
```

```
plt.show()
```



```
In [9]: import numpy as np

# Add jitter to reduce overlap
x = df['Internships'] + np.random.uniform(-0.1, 0.1, size=len(df))
y = df['Salary_LPA']

plt.figure(figsize=(8,5))

plt.scatter(x, y, alpha=0.5)

# Trend Line
z = np.polyfit(df['Internships'], df['Salary_LPA'], 1)
p = np.poly1d(z)
plt.plot(df['Internships'], p(df['Internships']))

plt.title("Internships vs Salary (Improved)")
plt.xlabel("Number of Internships")
plt.ylabel("Salary (LPA)")

plt.grid(True)

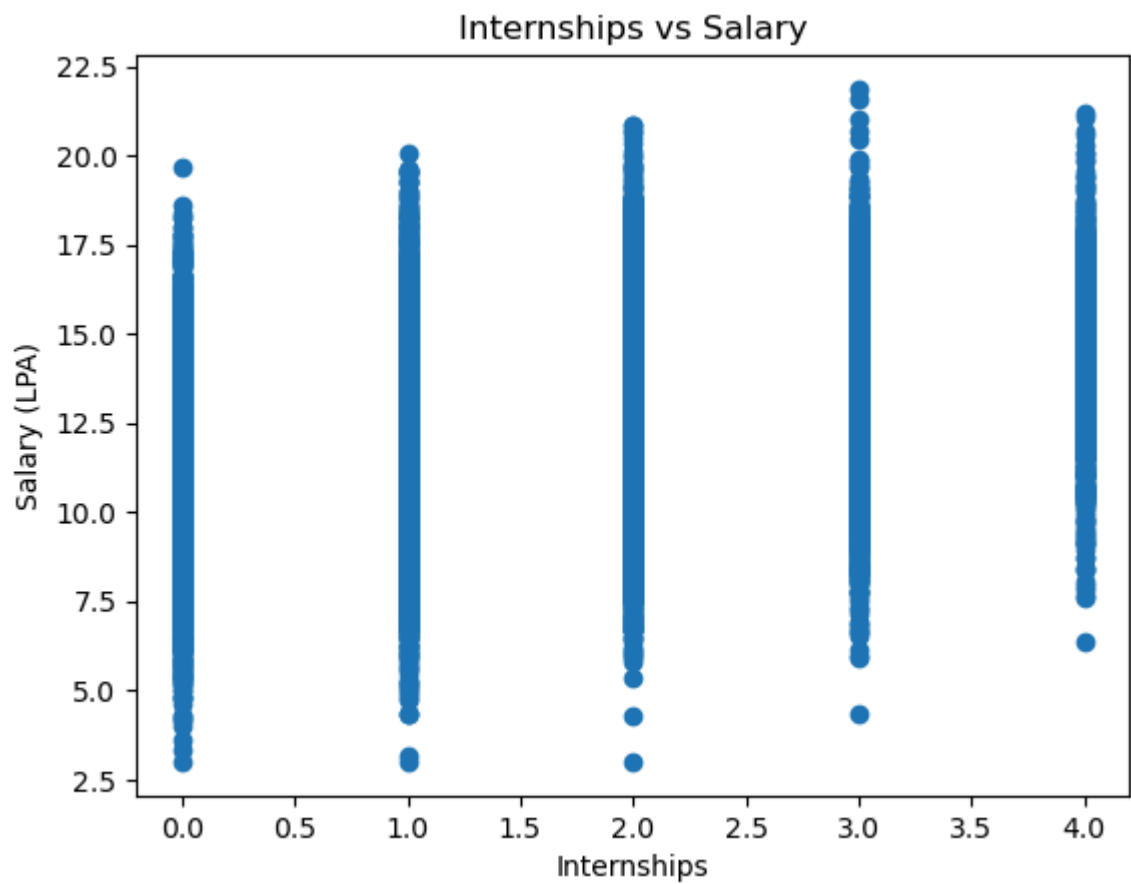
plt.show()
```



```
In [10]: plt.scatter(df['Internships'], df['Salary_LPA'])

plt.title("Internships vs Salary")
plt.xlabel("Internships")
plt.ylabel("Salary (LPA)")

plt.show()
```



```

In [11]: placement_counts = df['Placement_Status'].value_counts()

labels = ['Placed', 'Not Placed'] if 1 in placement_counts.index else placement_

plt.figure(figsize=(6,4))

bars = plt.bar(['Not Placed', 'Placed'], placement_counts)

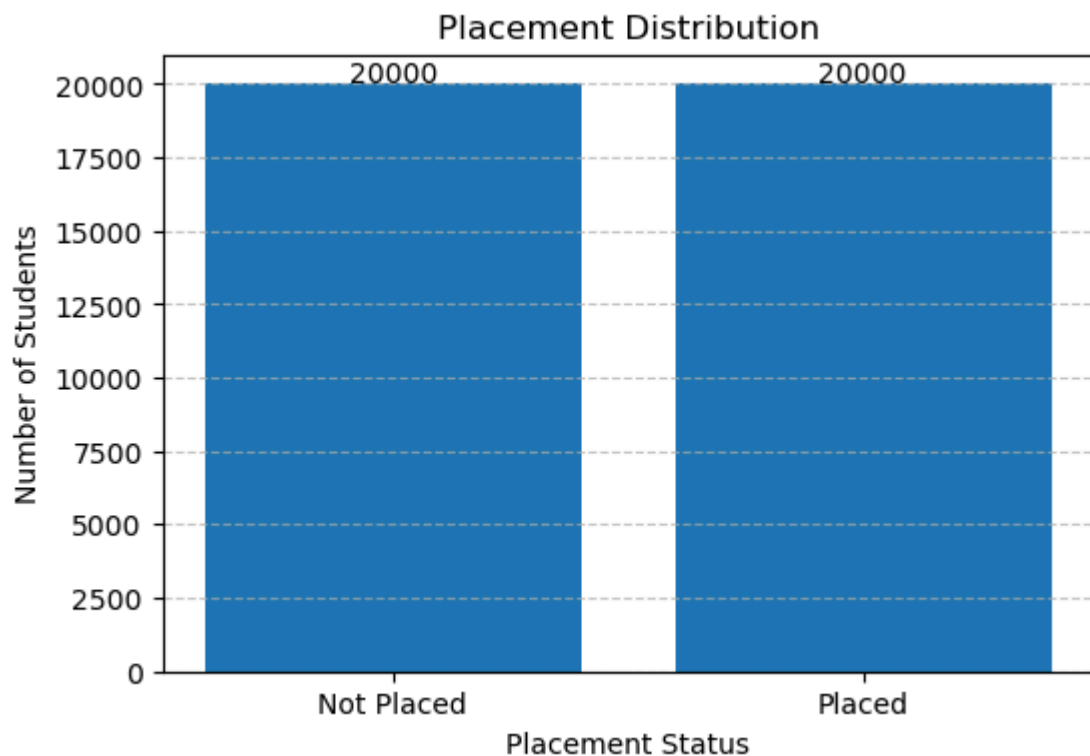
for bar in bars:
    yval = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, yval + 50, int(yval), ha='center')

plt.title("Placement Distribution")
plt.xlabel("Placement Status")
plt.ylabel("Number of Students")

plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()

```



```

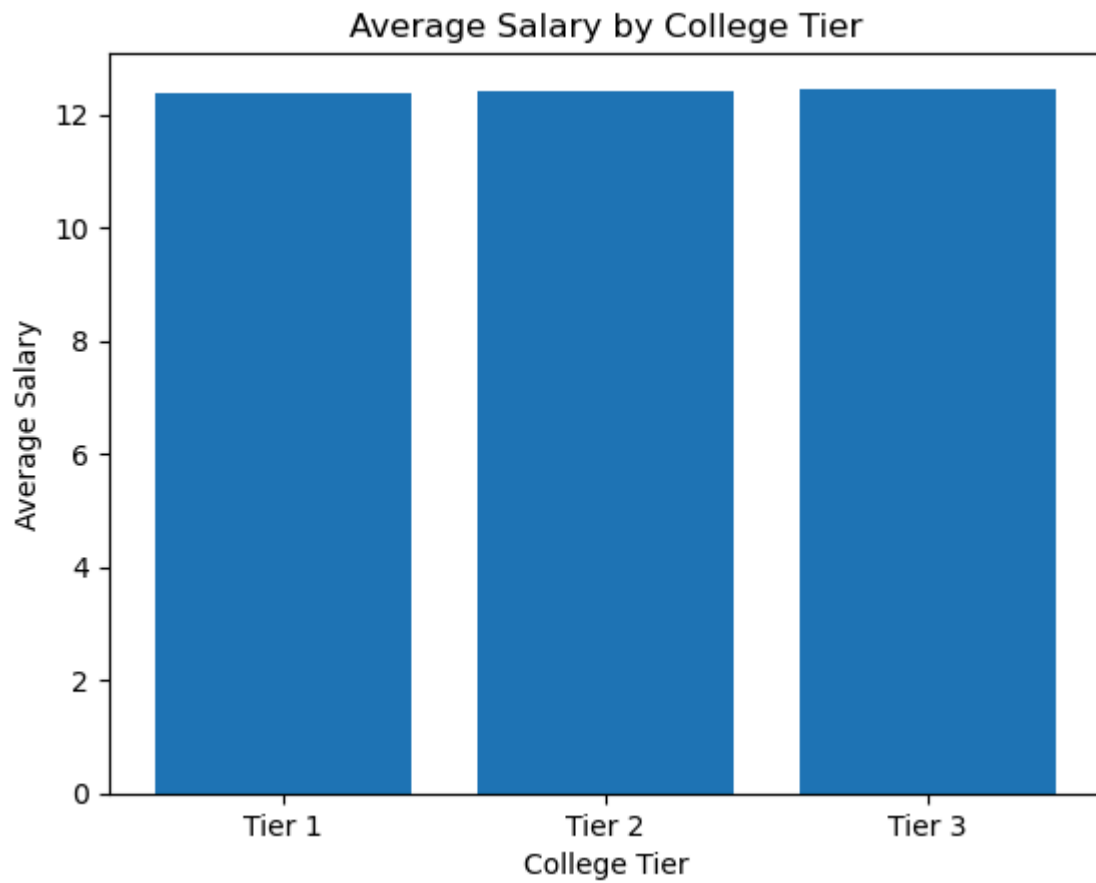
In [15]: avg_salary = df.groupby('College_Tier')['Salary_LPA'].mean()

plt.bar(avg_salary.index, avg_salary.values)

plt.title("Average Salary by College Tier")
plt.xlabel("College Tier")
plt.ylabel("Average Salary")

plt.show()

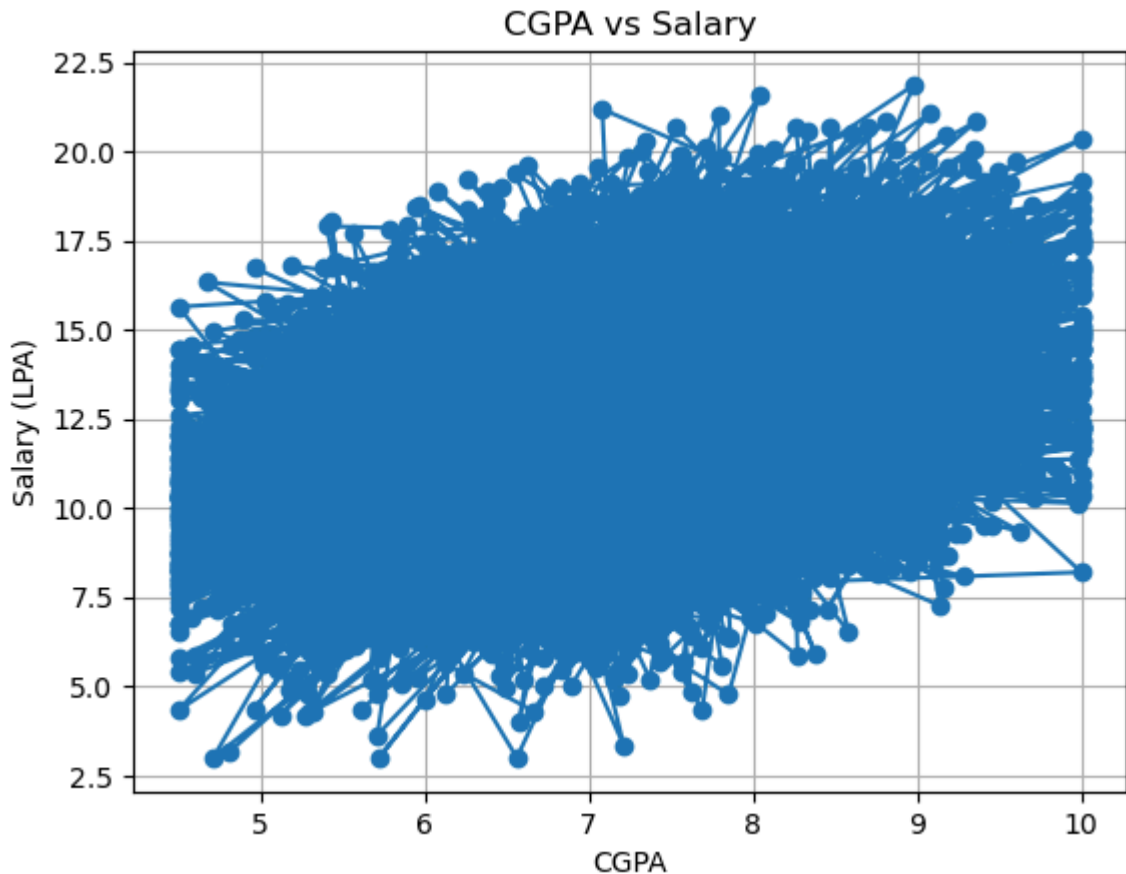
```



```
In [16]: plt.plot(df['CGPA'], df['Salary_LPA'], marker='o')

plt.title("CGPA vs Salary")
plt.xlabel("CGPA")
plt.ylabel("Salary (LPA)")
plt.grid()

plt.show()
```



```
In [16]: import pandas as pd
import matplotlib.pyplot as plt

# Create CGPA bins (VERY IMPORTANT)
df['CGPA_Bin'] = pd.cut(df['CGPA'], bins=[5,6,7,8,9,10])

# Calculate average salary per bin
avg_salary = df.groupby('CGPA_Bin')['Salary_LPA'].mean()

# Count students per bin (optional second line)
count_students = df.groupby('CGPA_Bin')['Salary_LPA'].count()

# -----
# PLOT CLEAN CHART
# -----
plt.figure(figsize=(10,6))

# Line 1 → Avg Salary
plt.plot(avg_salary.index.astype(str), avg_salary.values, marker='o', label='Avg')

# Line 2 → Student Count (insightful)
plt.plot(count_students.index.astype(str), count_students.values, marker='s', li

plt.title("CGPA vs Salary Trend (Binned Analysis)")
plt.xlabel("CGPA Range")
plt.ylabel("Values")
plt.legend()
plt.grid()

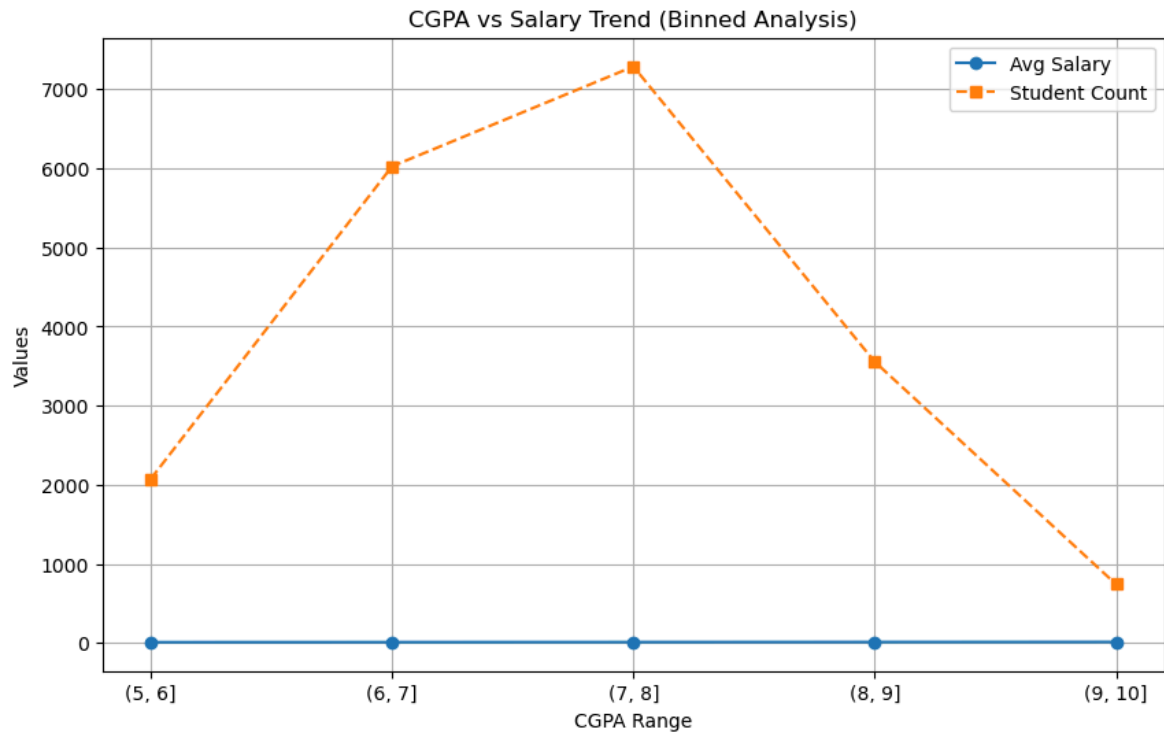
plt.show()
```

C:\Users\Vinay Kumar\AppData\Local\Temp\ipykernel_17920\938054798.py:8: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
avg_salary = df.groupby('CGPA_Bin')['Salary_LPA'].mean()
```

C:\Users\Vinay Kumar\AppData\Local\Temp\ipykernel_17920\938054798.py:11: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
count_students = df.groupby('CGPA_Bin')['Salary_LPA'].count()
```



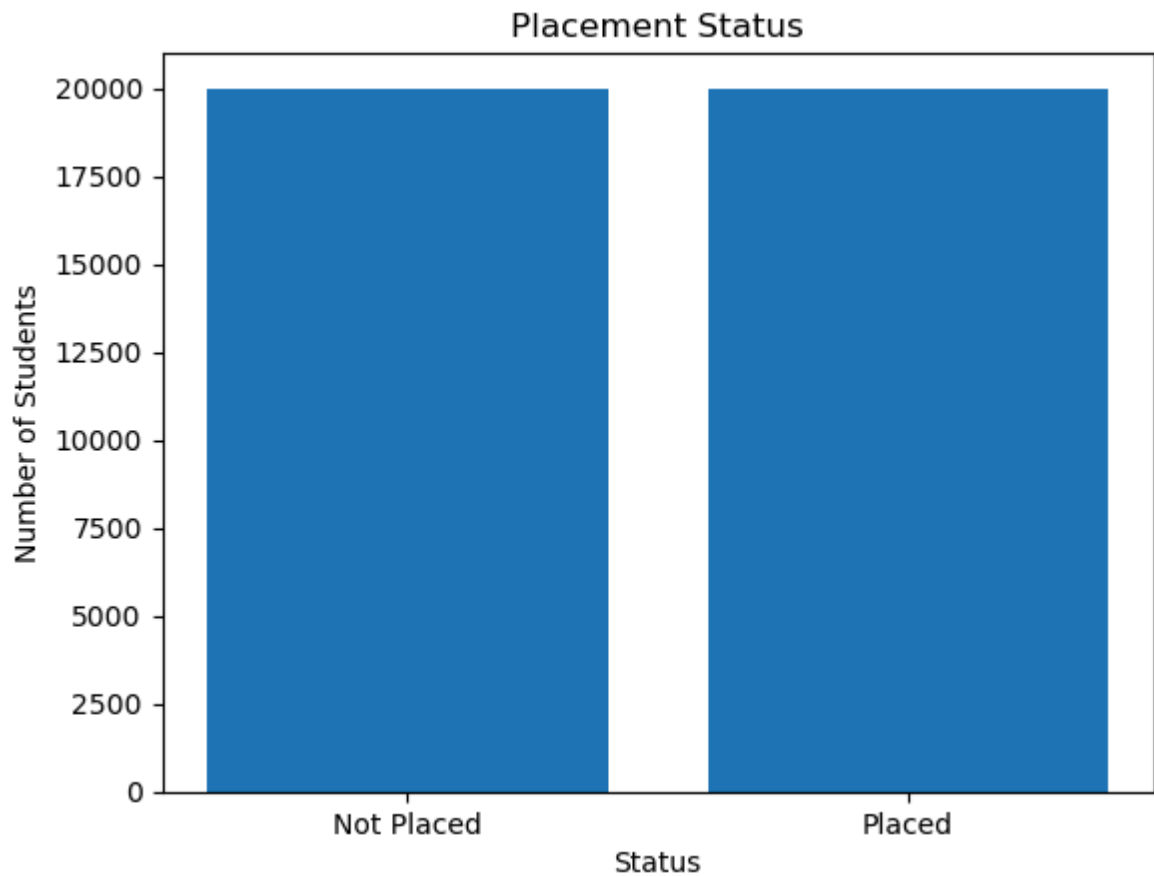
In []:

```
In [18]: placement_counts = df['Placement_Status'].value_counts()

plt.bar(['Not Placed', 'Placed'], placement_counts)

plt.title("Placement Status")
plt.xlabel("Status")
plt.ylabel("Number of Students")

plt.show()
```

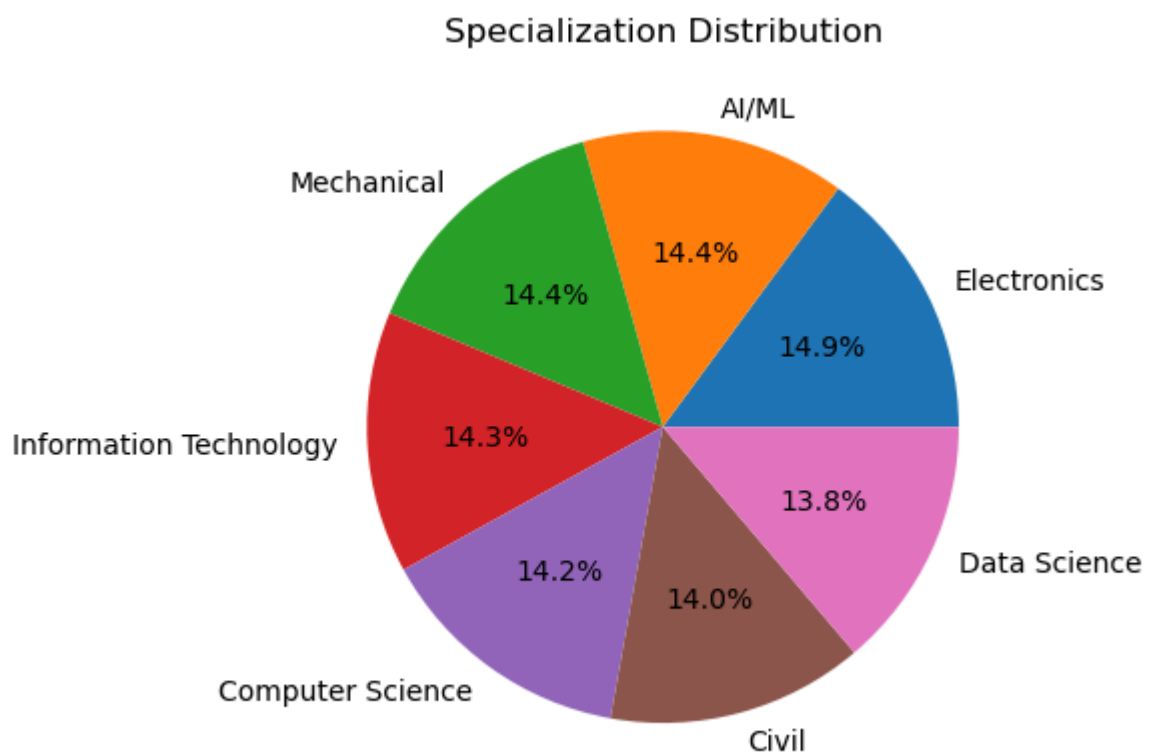


```
In [19]: spec_counts = df['Specialization'].value_counts()

plt.pie(spec_counts, labels=spec_counts.index, autopct='%1.1f%%')

plt.title("Specialization Distribution")

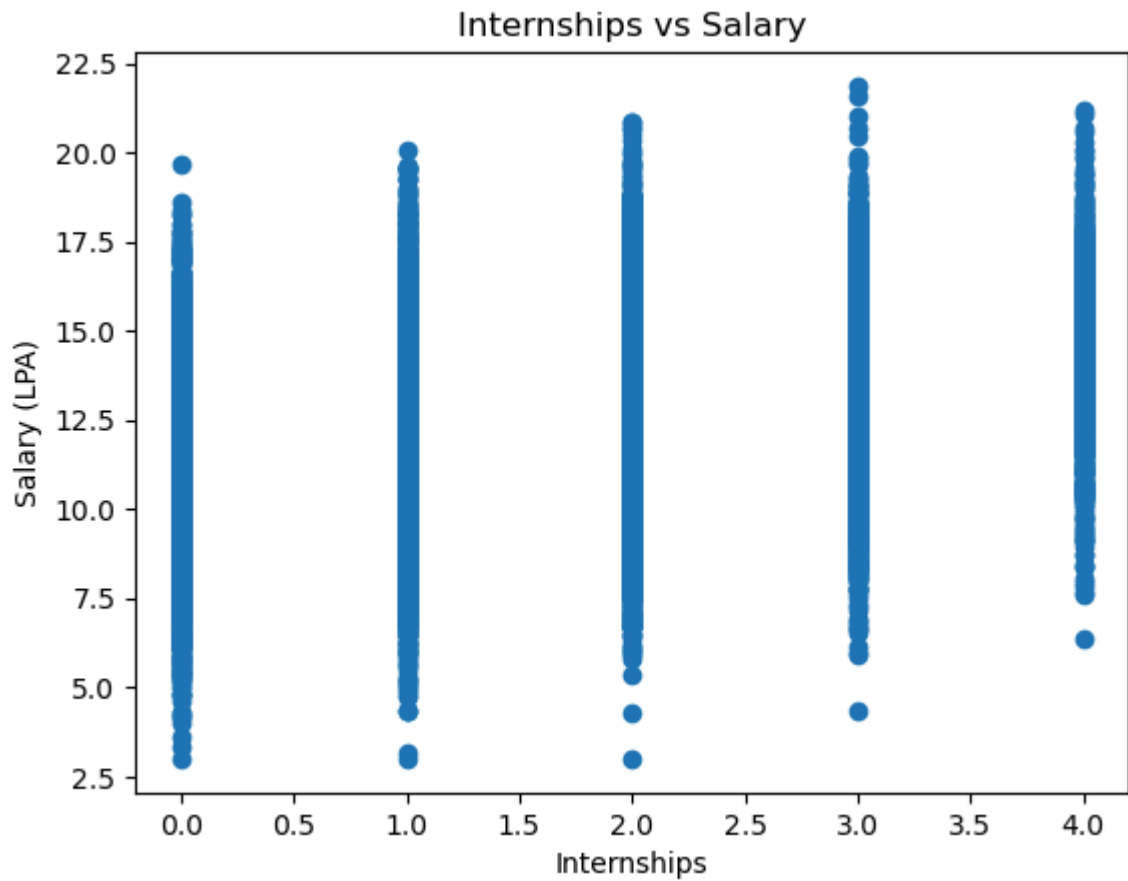
plt.show()
```




```
In [20]: plt.scatter(df['Internships'], df['Salary_LPA'])

plt.title("Internships vs Salary")
plt.xlabel("Internships")
plt.ylabel("Salary (LPA)")

plt.show()
```



```
In [21]: import pandas as pd

print(df.describe())

corr = df.corr(numeric_only=True)
print(corr)
```

	Age	CGPA	DSA_Problems_Solved	Internships \
count	20000.000000	20000.000000	20000.000000	20000.000000
mean	21.506600	7.202949	253.263700	1.450200
std	2.295347	1.005041	144.140893	1.160296
min	18.000000	4.500000	0.000000	0.000000
25%	19.000000	6.520000	150.000000	0.000000
50%	22.000000	7.210000	249.500000	1.000000
75%	24.000000	7.880000	352.000000	2.000000
max	25.000000	10.000000	841.000000	4.000000

	Certifications	Projects_Count	Communication_Skills \
count	20000.000000	20000.000000	20000.000000
mean	5.01255	6.503900	70.15475
std	3.14957	3.437932	17.70118
min	0.000000	1.000000	40.000000
25%	2.000000	3.000000	55.000000
50%	5.000000	7.000000	70.000000
75%	8.000000	9.000000	86.000000
max	10.000000	12.000000	100.000000

	Aptitude_Test_Score	LeetCode_Rating	GitHub_Contributions \
count	20000.000000	20000.000000	20000.000000
mean	67.488250	1452.676650	121.607400
std	18.995447	340.902149	74.906931
min	35.000000	800.000000	0.000000
25%	51.000000	1210.000000	65.000000
50%	67.000000	1447.000000	120.000000
75%	84.000000	1684.000000	173.000000
max	100.000000	2790.000000	478.000000

	Hackathons_Participated	AI_ML_Skill_Level	System_Design_Knowledge \
count	20000.000000	20000.000000	20000.000000
mean	3.993200	5.498350	5.490400
std	2.586559	2.867195	2.856522
min	0.000000	1.000000	1.000000
25%	2.000000	3.000000	3.000000
50%	4.000000	6.000000	5.000000
75%	6.000000	8.000000	8.000000
max	8.000000	10.000000	10.000000

	Resume_Score	Mock_Interview_Score	Placement_Status	Salary_LPA
count	20000.000000	20000.000000	20000.0	20000.000000
mean	72.318750	67.479050	1.0	12.433874
std	16.267538	19.031673	0.0	2.468236
min	45.000000	35.000000	1.0	3.000000
25%	58.000000	51.000000	1.0	10.760000
50%	72.000000	67.000000	1.0	12.430000
75%	87.000000	84.000000	1.0	14.080000
max	100.000000	100.000000	1.0	21.860000

	Age	CGPA	DSA_Problems_Solved	Internships \
Age	1.000000	-0.001615	0.004772	-0.003181
CGPA	-0.001615	1.000000	0.004095	-0.003204
DSA_Problems_Solved	0.004772	0.004095	1.000000	-0.003615
Internships	-0.003181	-0.003204	-0.003615	1.000000
Certifications	0.006300	-0.009704	0.000837	-0.008962
Projects_Count	-0.007531	0.001958	0.001448	0.004210
Communication_Skills	-0.001826	-0.007026	-0.010495	0.005535
Aptitude_Test_Score	-0.006489	0.000237	0.001696	-0.008547
LeetCode_Rating	-0.006466	0.003180	0.002069	-0.009320
GitHub_Contributions	0.010215	0.008865	-0.001925	0.000369

Hackathons_Participated	0.001347	0.002777	-0.006217	0.007551
AI_ML_Skill_Level	0.006631	0.001476	-0.001723	0.011676
System_Design_Knowledge	-0.007571	-0.011811	-0.000667	-0.003976
Resume_Score	0.003008	0.005258	-0.006103	0.002103
Mock_Interview_Score	0.007036	0.008567	-0.017437	0.005626
Placement_Status	NaN	NaN	NaN	NaN
Salary_LPA	0.005641	0.331696	0.003563	0.331986

	Certifications	Projects_Count	Communication_Skills	\
Age	0.006300	-0.007531	-0.001826	
CGPA	-0.009704	0.001958	-0.007026	
DSA_Problems_Solved	0.000837	0.001448	-0.010495	
Internships	-0.008962	0.004210	0.005535	
Certifications	1.000000	-0.002237	-0.003868	
Projects_Count	-0.002237	1.000000	-0.005939	
Communication_Skills	-0.003868	-0.005939	1.000000	
Aptitude_Test_Score	-0.001512	-0.000448	-0.000235	
LeetCode_Rating	-0.004271	-0.009044	0.008057	
GitHub_Contributions	-0.001896	-0.006804	0.015840	
Hackathons_Participated	0.001484	0.004614	0.004178	
AI_ML_Skill_Level	-0.008583	0.006141	-0.003468	
System_Design_Knowledge	-0.000968	0.012310	0.002959	
Resume_Score	0.004698	-0.007195	0.011278	
Mock_Interview_Score	-0.014372	-0.014755	0.011054	
Placement_Status	NaN	NaN	NaN	
Salary_LPA	-0.005066	0.014250	-0.000211	

	Aptitude_Test_Score	LeetCode_Rating	\
Age	-0.006489	-0.006466	
CGPA	0.000237	0.003180	
DSA_Problems_Solved	0.001696	0.002069	
Internships	-0.008547	-0.009320	
Certifications	-0.001512	-0.004271	
Projects_Count	-0.000448	-0.009044	
Communication_Skills	-0.000235	0.008057	
Aptitude_Test_Score	1.000000	0.001027	
LeetCode_Rating	0.001027	1.000000	
GitHub_Contributions	-0.009481	0.001225	
Hackathons_Participated	0.006700	-0.005727	
AI_ML_Skill_Level	0.012858	-0.001683	
System_Design_Knowledge	-0.003082	-0.000993	
Resume_Score	0.000386	-0.004662	
Mock_Interview_Score	0.002996	-0.007723	
Placement_Status	NaN	NaN	
Salary_LPA	0.006482	-0.000550	

	GitHub_Contributions	Hackathons_Participated	\
Age	0.010215	0.001347	
CGPA	0.008865	0.002777	
DSA_Problems_Solved	-0.001925	-0.006217	
Internships	0.000369	0.007551	
Certifications	-0.001896	0.001484	
Projects_Count	-0.006804	0.004614	
Communication_Skills	0.015840	0.004178	
Aptitude_Test_Score	-0.009481	0.006700	
LeetCode_Rating	0.001225	-0.005727	
GitHub_Contributions	1.000000	-0.000793	
Hackathons_Participated	-0.000793	1.000000	
AI_ML_Skill_Level	-0.002167	-0.004182	
System_Design_Knowledge	0.001356	-0.009984	

Resume_Score	0.005437	0.001986
Mock_Interview_Score	0.003477	0.007035
Placement_Status	NaN	NaN
Salary_LPA	0.003537	0.003808

	AI_ML_Skill_Level	System_Design_Knowledge	\
Age	0.006631	-0.007571	
CGPA	0.001476	-0.011811	
DSA_Problems_Solved	-0.001723	-0.000667	
Internships	0.011676	-0.003976	
Certifications	-0.008583	-0.000968	
Projects_Count	0.006141	0.012310	
Communication_Skills	-0.003468	0.002959	
Aptitude_Test_Score	0.012858	-0.003082	
LeetCode_Rating	-0.001683	-0.000993	
GitHub_Contributions	-0.002167	0.001356	
Hackathons_Participated	-0.004182	-0.009984	
AI_ML_Skill_Level	1.000000	-0.004095	
System_Design_Knowledge	-0.004095	1.000000	
Resume_Score	0.003570	-0.006927	
Mock_Interview_Score	-0.005125	0.005479	
Placement_Status	NaN	NaN	
Salary_LPA	0.363983	-0.005434	

	Resume_Score	Mock_Interview_Score	Placement_Status	\
Age	0.003008	0.007036	NaN	
CGPA	0.005258	0.008567	NaN	
DSA_Problems_Solved	-0.006103	-0.017437	NaN	
Internships	0.002103	0.005626	NaN	
Certifications	0.004698	-0.014372	NaN	
Projects_Count	-0.007195	-0.014755	NaN	
Communication_Skills	0.011278	0.011054	NaN	
Aptitude_Test_Score	0.000386	0.002996	NaN	
LeetCode_Rating	-0.004662	-0.007723	NaN	
GitHub_Contributions	0.005437	0.003477	NaN	
Hackathons_Participated	0.001986	0.007035	NaN	
AI_ML_Skill_Level	0.003570	-0.005125	NaN	
System_Design_Knowledge	-0.006927	0.005479	NaN	
Resume_Score	1.000000	-0.001155	NaN	
Mock_Interview_Score	-0.001155	1.000000	NaN	
Placement_Status	NaN	NaN	NaN	
Salary_LPA	0.006448	0.002738	NaN	

	Salary_LPA
Age	0.005641
CGPA	0.331696
DSA_Problems_Solved	0.003563
Internships	0.331986
Certifications	-0.005066
Projects_Count	0.014250
Communication_Skills	-0.000211
Aptitude_Test_Score	0.006482
LeetCode_Rating	-0.000550
GitHub_Contributions	0.003537
Hackathons_Participated	0.003808
AI_ML_Skill_Level	0.363983
System_Design_Knowledge	-0.005434
Resume_Score	0.006448
Mock_Interview_Score	0.002738

```
Placement_Status      NaN
Salary_LPA            1.000000
```

```
In [22]: import matplotlib.pyplot as plt

corr = df.corr(numeric_only=True)

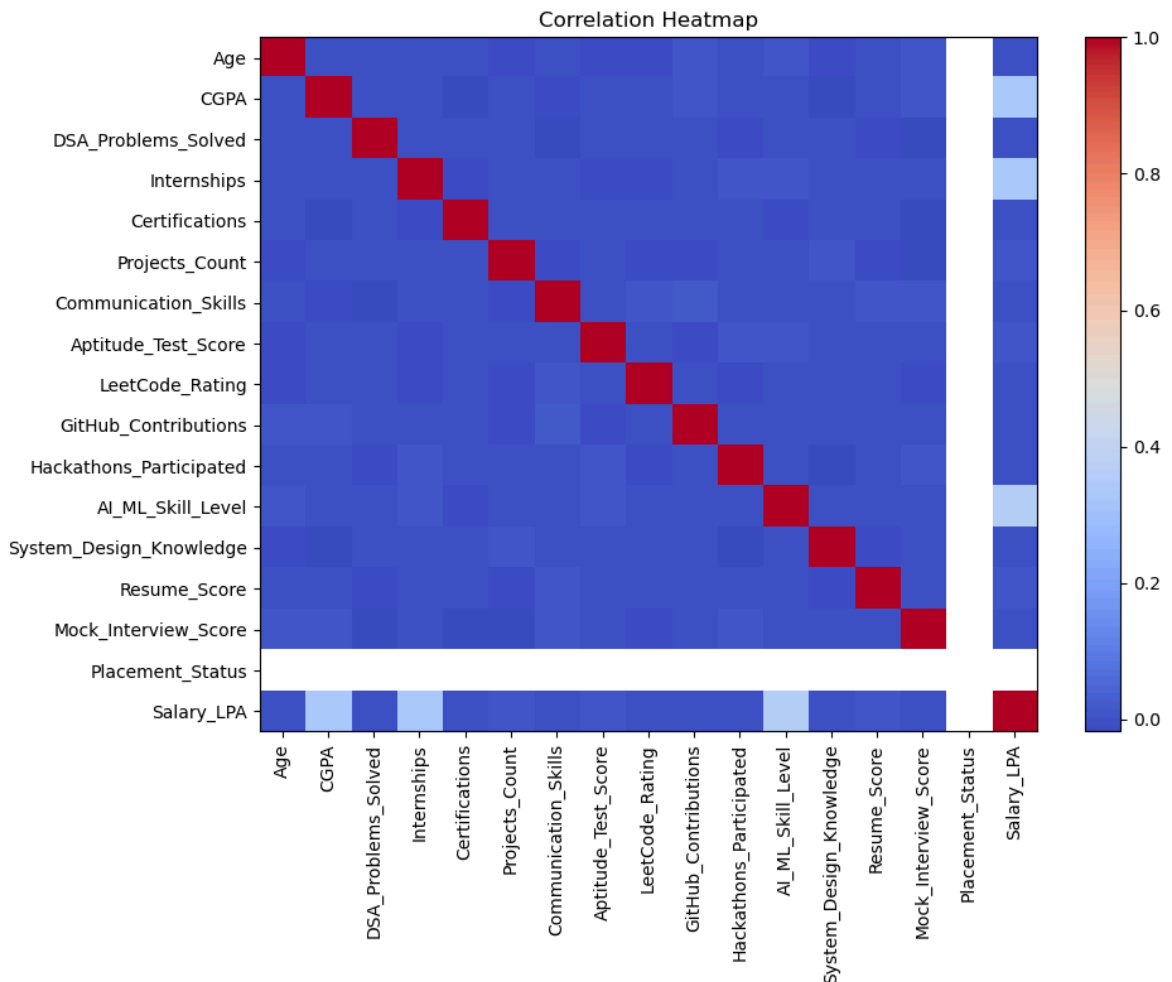
plt.figure(figsize=(10,8))
plt.imshow(corr, cmap='coolwarm', aspect='auto')

plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)

plt.title("Correlation Heatmap")

plt.tight_layout()
plt.show()
```

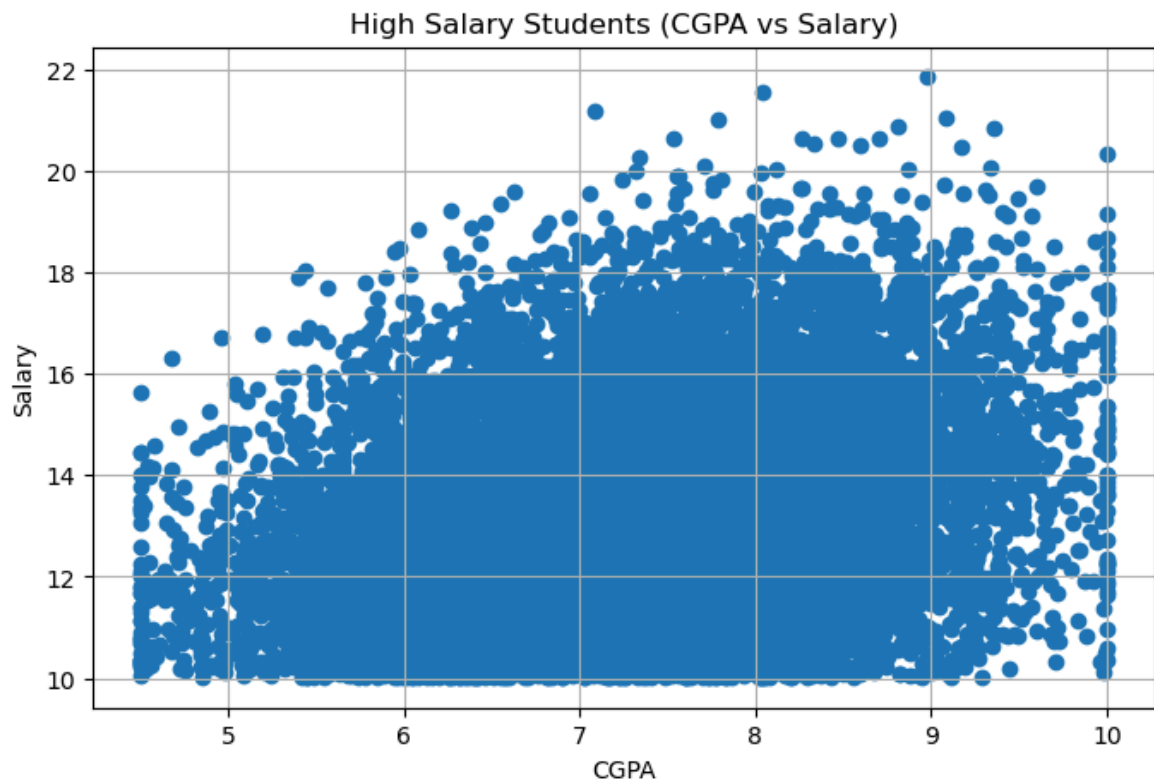


```
In [23]: high_salary = df[df['Salary_LPA'] > 10]

plt.figure(figsize=(8,5))
plt.scatter(high_salary['CGPA'], high_salary['Salary_LPA'])

plt.title("High Salary Students (CGPA vs Salary)")
plt.xlabel("CGPA")
plt.ylabel("Salary")
```

```
plt.grid()
plt.show()
```



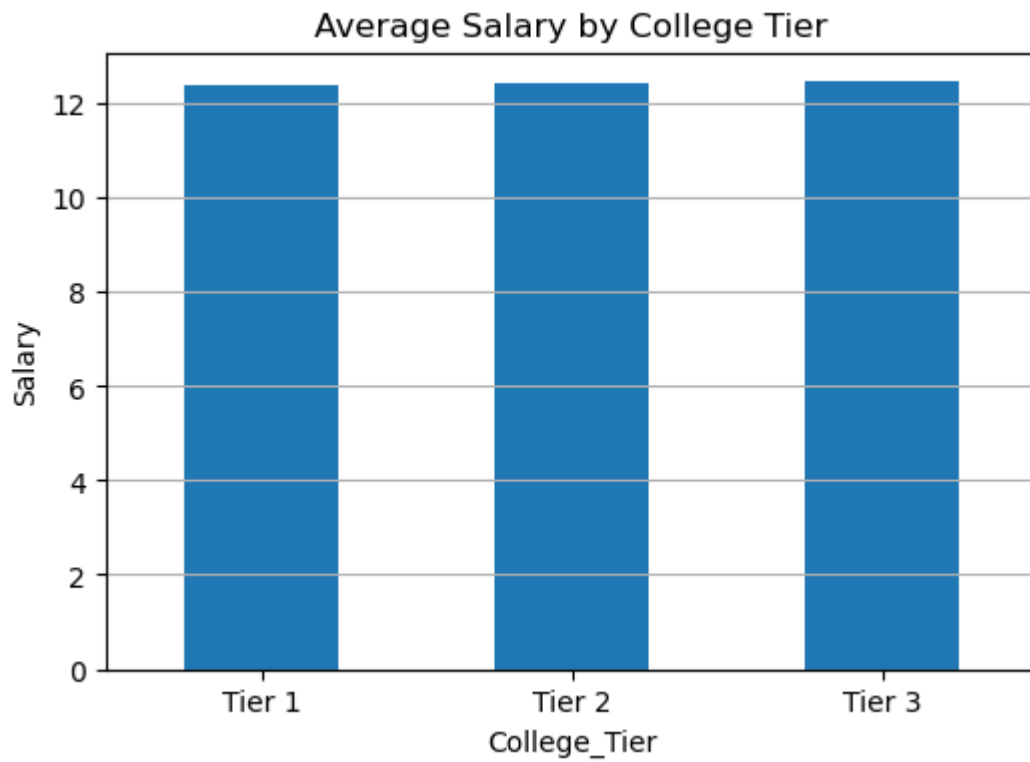
```
In [24]: avg_salary = df.groupby('College_Tier')['Salary_LPA'].mean().sort_values()

plt.figure(figsize=(6,4))
avg_salary.plot(kind='bar')

plt.title("Average Salary by College Tier")
plt.ylabel("Salary")

plt.xticks(rotation=0)
plt.grid(axis='y')

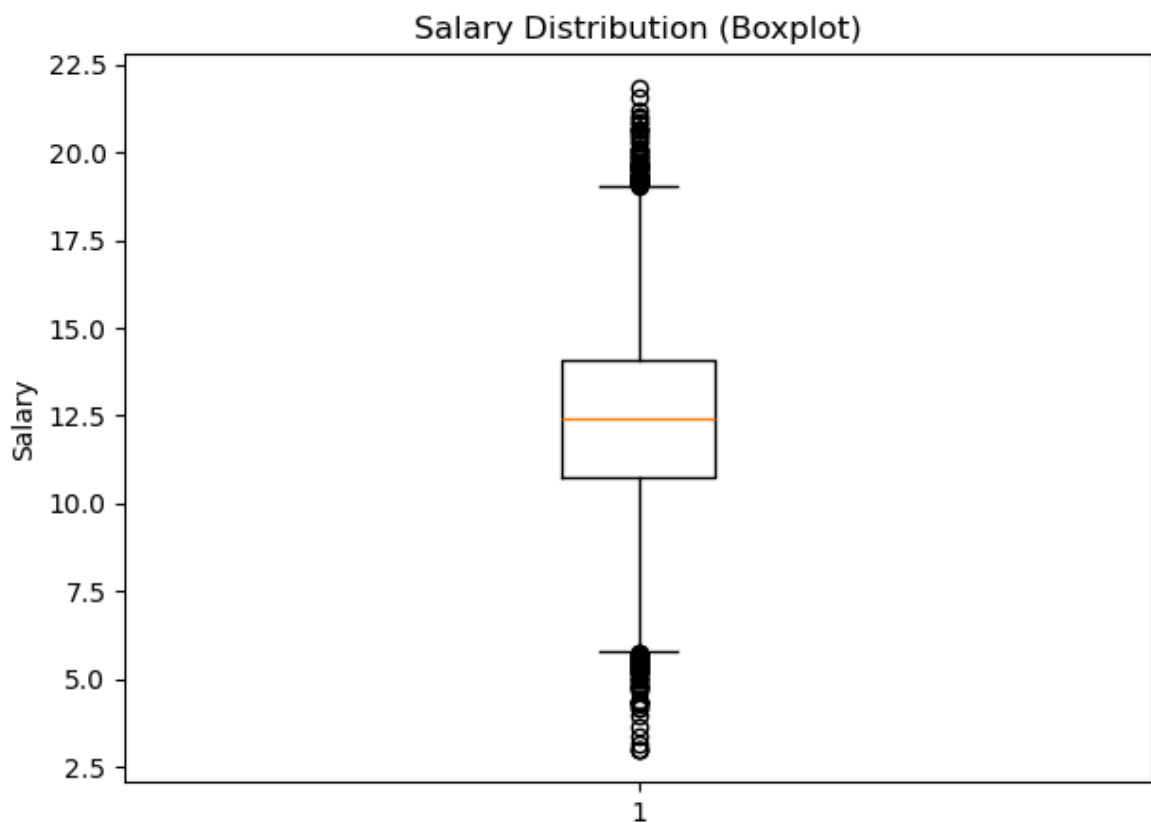
plt.show()
```



```
In [25]: plt.figure(figsize=(7,5))
plt.boxplot(df['Salary_LPA'])

plt.title("Salary Distribution (Boxplot)")
plt.ylabel("Salary")

plt.show()
```



```
In [21]: import pandas as pd
import matplotlib.pyplot as plt
```

```

import numpy as np

# Load data
df = pd.read_csv(r"C:\Users\Vinay Kumar\Downloads\archive (3)\student_placement_

# -----
# CREATE FIGURE (DASHBOARD)
# -----
fig = plt.figure(figsize=(16,10))
fig.suptitle(" Student Placement Dashboard", fontsize=20)

# -----
# 1. CGPA vs Salary (Scatter + Trend)
# -----
ax1 = plt.subplot(2,3,1)

x = df['CGPA']
y = df['Salary_LPA']

ax1.scatter(x, y, alpha=0.5)

# Trend Line
z = np.polyfit(x, y, 1)
p = np.poly1d(z)
ax1.plot(x, p(x))

ax1.set_title("CGPA vs Salary")
ax1.set_xlabel("CGPA")
ax1.set_ylabel("Salary")
ax1.grid()

# -----
# 2. Avg Salary vs Internships
# -----
ax2 = plt.subplot(2,3,2)

avg_salary = df.groupby('Internships')['Salary_LPA'].mean()

ax2.plot(avg_salary.index, avg_salary.values, marker='o')

ax2.set_title("Avg Salary vs Internships")
ax2.set_xlabel("Internships")
ax2.set_ylabel("Avg Salary")
ax2.grid()

# -----
# 3. Salary Distribution
# -----
ax3 = plt.subplot(2,3,3)

ax3.hist(df['Salary_LPA'])

ax3.set_title("Salary Distribution")
ax3.set_xlabel("Salary")
ax3.set_ylabel("Frequency")

# -----
# 4. Placement vs Salary (Box Plot)
# -----

```



```

ax4 = plt.subplot(2,3,4)

# Create Labels
df['Placement_Label'] = df['Placement_Status'].map({1: 'Placed', 0: 'Not Placed'})

# Boxplot
ax4.boxplot(
    [df[df['Placement_Label']=='Placed']['Salary_LPA'],
     df[df['Placement_Label']=='Not Placed']['Salary_LPA']],
    labels=['Placed', 'Not Placed'])

ax4.set_title("Salary Distribution by Placement Status")
ax4.set_xlabel("Placement Status")
ax4.set_ylabel("Salary (LPA)")
ax4.grid(axis='y')

# 5. Avg Salary by College Tie
# -----
ax5 = plt.subplot(2,3,5)

tier_salary = df.groupby('College_Tier')['Salary_LPA'].mean()

ax5.bar(tier_salary.index, tier_salary.values)

ax5.set_title("Avg Salary by College Tier")
ax5.set_xlabel("College Tier")
ax5.set_ylabel("Salary")
ax5.grid(axis='y')

# -----
ax6 = plt.subplot(2,3,6)

spec_counts = df['Specialization'].value_counts()

plt.pie(spec_counts, labels=spec_counts.index, autopct='%1.1f%%')

plt.title("Specialization Distribution")

plt.show()

# -----
# FINAL SPACING (VERY IMPORTANT)
# -----
plt.tight_layout(rect=[0, 0, 1, 0.96])

plt.show()

```

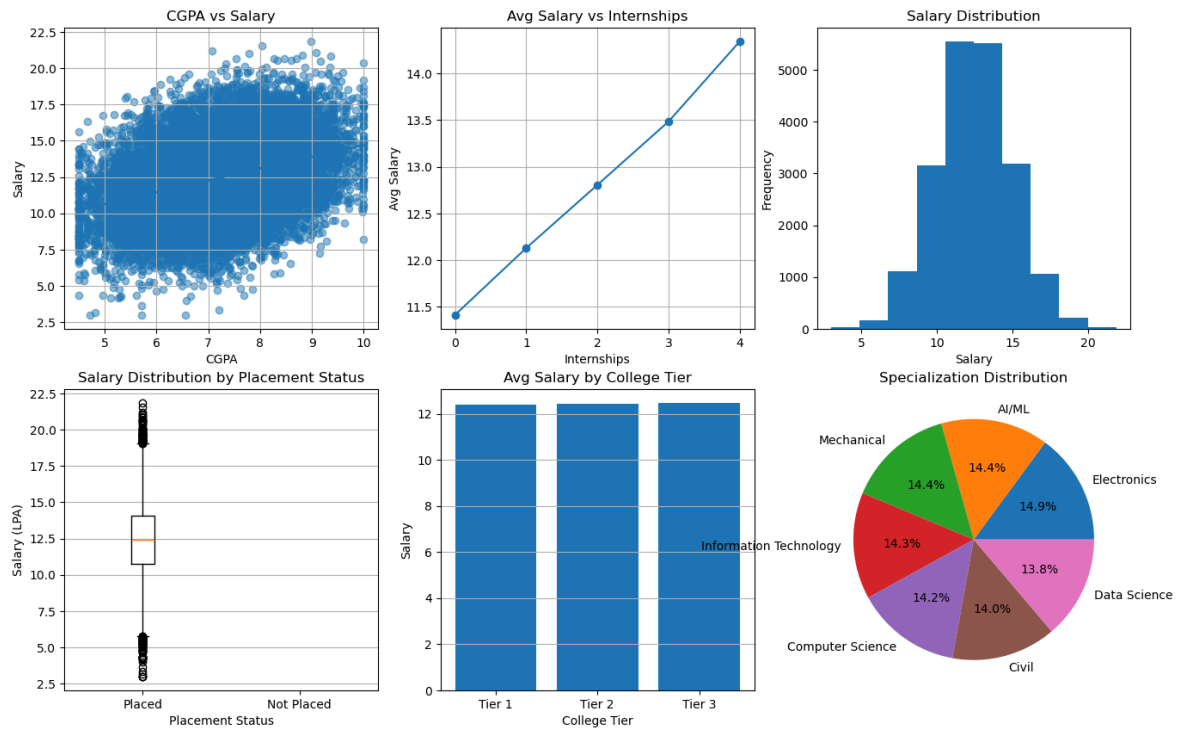
C:\Users\Vinay Kumar\AppData\Local\Temp\ipykernel_22740\3904529659.py:69: MatplotlibDeprecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_labels' since Matplotlib 3.9; support for the old name will be dropped in 3.11.

```

ax4.boxplot(

```

Student Placement Dashboard



<Figure size 640x480 with 0 Axes>

In []: